Robust Z Scores

In order to explore how the choice of different features affects flags for the baseline model, I compared 3 sets of features. Everything else stayed the same: cleaned data, peer groups (minimum size of 100), and the flagging rule. I also classified raw features into two categories: volume (patients, services, service days) and price (submitted charge, allowed amount, payment, standardized payment).

Feature set	Features
A: all 7 raw	Tot_Benes, Tot_Srvcs, Tot_Bene_Day_Srvcs, Avg_Sbmtd_Chrg, Avg_Mdcr_Alowd_Amt, Avg_Mdcr_Pymt_Amt, Avg_Mdcr_Stdzd_Amt
B: 4 raw	Tot_Benes, Tot_Srvcs, Avg_Sbmtd_Chrg, Avg_Mdcr_Stdzd_Amt
C: ours	standardized payment, beneficiaries, services per beneficiary, markup

Table of feature sets

The main problem with correlated features is that the model can flag the same reason twice or more. Since anomalies are found based on the rule (1 feature > 10 or 2+ features > 3.5), each feature should carry information that is not already represented by other columns. I counted a flag as double-counted if it was caused by 2 or more features > 3.5, no features > 10, and all of them belonged to the same category.

feature set	n features	flag %	double-counted % of flags
A: all 7 raw	7	0.99	70.75
B: 4 raw	4	0.60	53.89
C: ours (2 raw + 2 FE)	4	1.24	0.00

Table 1: Flag rates and double-counted flags

Results from the baseline model showed that feature set C flags more rows (1.24%) than sets A (0.99%) and B (0.60%). At the same time, for sets A and B more than half of the flags are double-counted (71% and 54%). This means that the model often counts the same reason multiple times. Set C has 0% double-counted flags under this definition, since each feature is treated as a separate category.

	A: all 7 raw	B: 4 raw	C: ours (2 raw + 2 FE)
A: all 7 raw	100.0	66.8	28.1
B: 4 raw	66.8	100.0	28.0
C: ours (2 raw + 2 FE)	28.1	28.0	100.0

Table 2: Overlap of the top 1% between feature sets

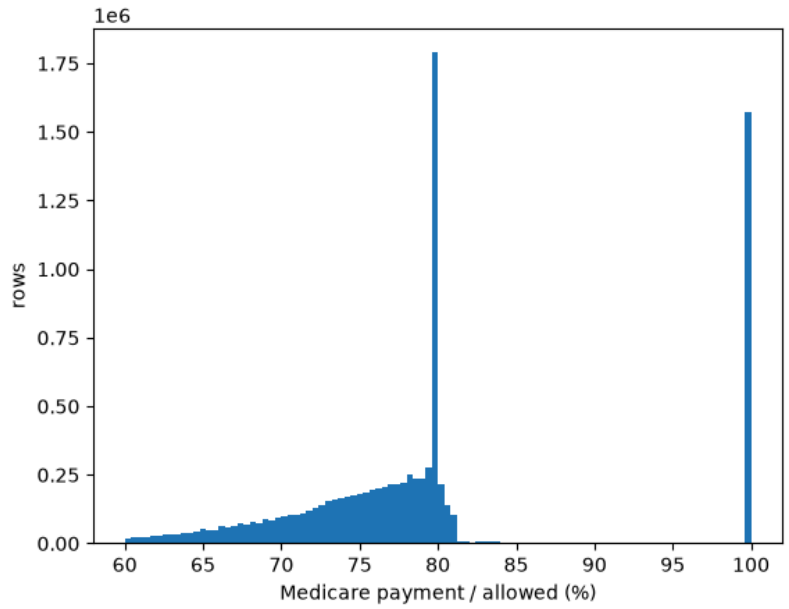
The overlap between set C and sets A and B is really small, just around 28%. This means that 72% of the top anomalies found with our engineered features were not in the top 1% of the sets with 7 or 4 raw features when compared separately.

For example, some rows flagged only by raw features are bigger practices where patients, services, and service days are all slightly higher than usual. In this case, the model counts basically the same information 3 times. Another example is providers in high-cost areas where the allowed amount is high, but standardized payment is normal.

Our engineered features help avoid this problem because each feature represents something different: standardized payment, number of beneficiaries, services per beneficiary, and markup. The correlation analysis also showed that these features have low correlations with each other.

Checking payment_to_allowed

To check whether payment_to_allowed adds useful information, I added it to set C as a fifth feature. It created 51,306 new flags (29.7% of all flags). In 86% of these cases, payment_to_allowed alone had a robust z-score > 10, and in 75% the other four features were normal. At the same time, the median difference from the peer median was only 0.8 percentage points.



Histogram of payment_to_allowed

The problem is that this value mostly depends on Medicare rules: around 80% for most services, 100% for services without patient coinsurance (16.2% of rows), and lower when the patient's deductible applies. Within peer groups, the value barely changes. The median IQR is only 0.08 percentage points, and for 25% of groups it is 0. Because of this, even small differences can result in very high z-scores.

This might also affect Isolation Forest, since it can isolate values that are only slightly different from a large group of similar values. So I don't think payment_to_allowed is a good feature for our model, as these differences are more likely related to Medicare rules rather than unusual provider behavior.

Isolation Forest

New columns added:
1) services_per_bene - total services/total beneficiaries
2) services_per_bene_day - total services/total beneficiaries-day services
3) charge_to_allowed - average submitted charge/average allowed amount by medicare
4) payment_to_allowed - average medicare payment amount/average medicare allowed amount

Methods used:

Within peer feature variation:
Why is it used: isolation forest can only use a feature to distinguish records if that feature actually varies within the population being modeled Global variation across the entire Medicare dataset is not sufficient because isolation forest is trained separately within individual peer groups. Therefore the variation was measured within the peer group rather than only across the full dataset.
Analysis examined:

- Median within peer standard deviation
- Median within peer interquartile range
- Median number of unique values
- Percentage of peer groups where the feature was constant

Results:

feature	median_peer_std	median_peer_iqr	median_unique_values	pct_constant_peers
Tot_Srvcs	0.769592	1.059222	108.0	0.0
Tot_Bene_Day_Srvcs	0.769257	1.041363	102.5	0.0
Tot_Benes	0.705747	1.002773	84.0	0.0
Avg_Sbmtd_Chrg	0.512390	0.617051	224.0	0.0
charge_to_allowed	0.386886	0.430149	315.0	0.0
Avg_Mdcr_Pymt_Amt	0.157952	0.139548	313.5	0.0
Avg_Mdcr_Alowd_Amt	0.118460	0.104189	313.5	0.0
Avg_Mdcr_Stdzd_Amt	0.111589	0.088576	281.0	6.0
services_per_bene	0.062990	0.059233	137.5	0.0
payment_to_allowed	0.021196	0.026622	266.5	14.0
services_per_bene_day	0.002516	0.000000	7.0	22.0

Interpretation:
The clearest concern is services_per_bene_day. Its median within peer IQR was essentially zero, it had only seven unique values in the median peer group, and it was completely constant in approximately 22% of the tested peers. This means that in many peer groups it provides little or no ability for an isolated forest to distinguish one observation from another.

Within peer spearman correlation
Why it was used: strongly correlated features can contain largely overlapping information. This is particularly important for isolation forest because redundant variables can effectively give one behavioral dimension opportunities to influence random tree splits. Spearman correlation was used rather than only relying on pearson correlation because the medicare variables are highly skewed and the relationships between variables do not need to be linear. Correlations were calculated within peer groups, and the median correlation across peer groups was examined. A correlation greater than 0.85 was treated as a redundancy warning.

Results:

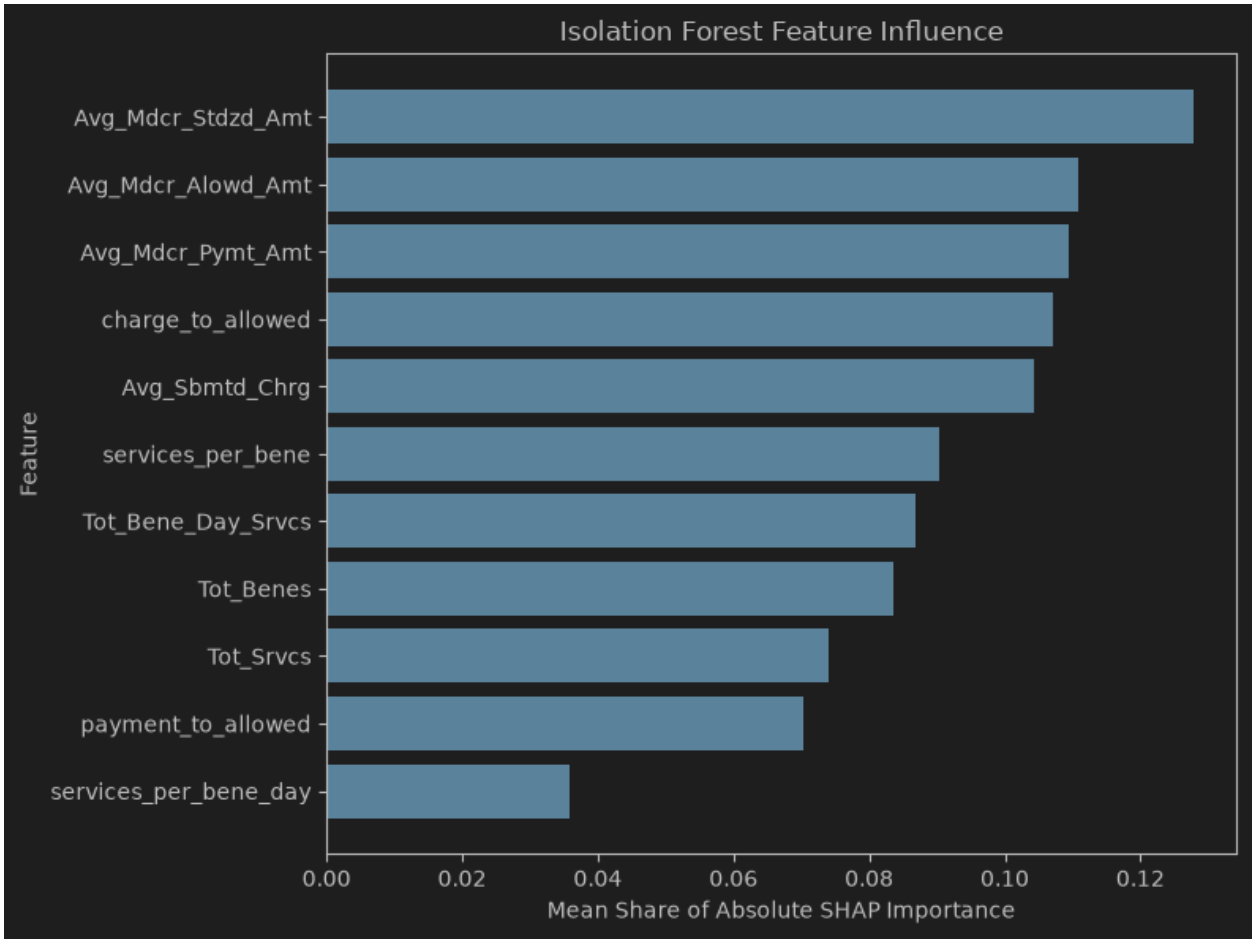
	feature_1	feature_2	median_abs_spearman
2	Tot_Srvcs	Tot_Bene_Day_Srvcs	0.999958
1	Tot_Benes	Tot_Bene_Day_Srvcs	0.994135
0	Tot_Benes	Tot_Srvcs	0.984645
3	Avg_Sbmtd_Chrg	charge_to_allowed	0.954078
4	Avg_Mdcr_Alowd_Amt	Avg_Mdcr_Pymt_Amt	0.941129

Interpretation:
Tot_Benes, Tot_Srvcs, and Tot_Bene_Day_Services form an extremely strong redundancy cluster. Their correlations range from approximately 0.985 to almost 1.00. Therefore, including all these three would provide very little new information while repeatedly representing the same general volume dimension. Tot_Benes remains a reasonable representative because it has a clear interpretation.

Avg_Sbmtd_Chrg and charge_to_allowed have a correlation of approximately 0.95. This was expected because the submitted charge is mathematically included in the charge to allowed ratio.
Avg_Mdcr_Alowd_Amnt and Avg_Mdcr_Pymt_Amt had a correlation of approximately 0.94. This suggests that including both raw payment measures would likely add a considerable redundancy.

SHAP feature influence
Why it is used: SHAP was used to examine which features the fitted isolation forest relied on when producing anomaly scores. SHAP measures how strongly the existing model uses a feature

Results:



Interpretation:
Avg_Stdzd_Amt had the highest average SHAP influence, suggesting that standardized Medicare payment information contributes meaningfully to the model's anomaly scores. At the opposite end, services_per_bene_day had the lowest average influence.

Leave one feature out ablation
Why it is used:
Ablation provides stronger evidence whether a feature is necessary. The full candidate model was first trained using all candidate features. Then the model was repeatedly retrained while removing exactly one feature at a time. Each reduced model was evaluated on the same records as the full model. Two measures were calculated:

Spearman rank correlation:
Spearman correlation measured agreement between the entire anomaly ranking produced by the full model and the ranking produced after removing one feature
Interpretation: closer to 1 = removing the feature changed very little. Lower value = removing the feature materially changed the ranking.

Top 5% jaccard similarity:
The jaccard score measured overlap between the most anomalous 5% of records from the full model and the reduced model.
Interpretation: closer to 1 = essentially the same review queue. Lower = removing the feature changed which records appear among the strongest anomalies.

removed_feature	mean_spearman	median_spearman	std_spearman	mean_top5_jaccard	median_top5_jaccard
services_per_bene	0.947906	0.955424	0.030487	0.724548	0.714286
Avg_Sbmtd_Chrg	0.957847	0.959432	0.016228	0.742836	0.782506
payment_to_allowed	0.960335	0.965233	0.021066	0.747559	0.777778
charge_to_allowed	0.960961	0.961228	0.017055	0.730876	0.740230
Avg_Mdcr_Alowd_Amt	0.967368	0.970226	0.013792	0.752407	0.777778
Avg_Mdcr_Stdzd_Amt	0.970153	0.972101	0.013730	0.688023	0.678699
Avg_Mdcr_Pymt_Amt	0.971151	0.973468	0.013315	0.709048	0.706458
services_per_bene_day	0.973392	0.977470	0.017911	0.751869	0.755556
Tot_Bene_Day_Srvcs	0.975939	0.978326	0.010851	0.783629	0.800000
Tot_Benes	0.975942	0.979282	0.012082	0.825510	0.955696
Tot_Srvcs	0.976572	0.978895	0.009905	0.791443	0.835586

Major findings:
Removing services_per_bene produced the lowest overall spearman correlation: 0.9479. This means its removal changed the complete anomaly ranking more than the removal of any other tested feature.

Services_per_bene_day showed several weaknesses simultaneously:

- Median within peer IQR about 0
- Median unique values = 7
- Lowest SHAP importance
- Removal still produces a relatively high Spearman correlation of 0.9734

Services_per_bene_day is strongest candidate for removal

Avg_Mdcr_Stdzd_amt produces one of the most interesting results. It has the highest average SHAP importance. Its removal did not dramatically change the complete ranking. However, removing produces the lowest top 5% jaccard score of all features. Therefore, standardized payment appears particularly important for determining which observations reach the extreme end of the anomaly ranking.

Three volume measures Tot_Benes, Tot_Srvc, and Tot_Bene_Day_Srvcs have a correlation between 0.985 and 1.00 making them excessively redundant. Ablation results were also very similar. Tot_benes is a reasonable representative because it is directly interpretable as provider-service beneficiary volume.

Retain:
Services per beneficiary
Charge / allowed
Total beneficiaries
Payment / allowed
Average medicare standardized amount

DBSCAN

Columns Used:
HCPCS Code (1),
Place of Service (2),
Provider Type (3),
Provider Entity Code (4),
State Abbreviation (5)

Ran DBSCAN on these combinations of those columns:
12, 123, 1234, 1235, 1245, 12345

Main Findings:
12: Broad baseline, but groups are big and doesn't provide detailed enough groups to reasonably explain outliers
123: Provider type really splits the groups and makes each group detailed enough to reasonably explain how the outliers are anomalies.
1234: Nearly the same as 123, but the splits made aren't that useful in information gained. More blurs how useful the groups are instead of adding more detail.
1235: Adds a good amount of context via the Geolocation, however the groups are way smaller and most aren't big enough to analyze accurately.
12345: nearly the same as 1235, but the same issue as before where 4 will cause problems with the groups being needlessly split.
1245: About the same as 1235. However it causes problem since group 3 adds required context and the groups can be made to show outliers that realistically don't make sense.

Overall Recommendations:
123: For accurate analysis among most data.
1235: For analysis with geolocation.